

An Analytic Hamiltonian KAM Theorem

Statement of the Current Lean Interface

Revised English formulation | October 8, 2026

This note states the mathematical input and output of `KamProject.Arnold1963.theorem1`. It presents the project's basic nondegenerate analytic Hamiltonian KAM theorem, developed with reference to Arnold (1963), under the explicit assumptions of the current code. The source version and changes from the earlier English draft are recorded on page 4.

1 Fixed data, domains, and conventions

Fix an integer $n \geq 1$. Identify $\mathbb{R}^n$ with its coordinatewise inclusion in $\mathbb{C}^n$, and use the coordinate maximum norm on both spaces. The norm on $\mathbb{C}^n \times \mathbb{C}^n$ is the product maximum norm. Coordinates are ordered as (p, q) , with p the action and q the angle.

Let $G \subset \mathbb{C}^n$ be compact and invariant under complex conjugation. Define its real slice by

$$G_{\mathbb{R}} := \{p \in \mathbb{R}^n : p \in G\}.$$

Assume that a nonempty open set $U \subset \mathbb{R}^n$ is specified such that

$$\overline{U} = G_{\mathbb{R}}, \quad m_n(\partial U) = 0, \quad U \subset \text{int}_{\mathbb{C}^n} G, \quad (1)$$

where closures and boundaries of real sets are taken in $\mathbb{R}^n$, and m_n is Lebesgue measure. In the last inclusion, real points are viewed as complex points. These assumptions imply that U is bounded and that $0 < m_n(G_{\mathbb{R}}) < \infty$. No complex-interior condition is imposed on every point of $\overline{U}$.

Let $H_0 : \mathbb{C}^n \rightarrow \mathbb{C}$ be holomorphic on a neighborhood of G , and assume

$$H_0(\overline{p}) = \overline{H_0(p)} \quad (p \in G), \quad \det D^2 H_0(p) \neq 0 \quad (p \in U). \quad (2)$$

The Hessian uses complex coordinate derivatives. Its determinant is required to be nonzero only on U ; the frequency map $A = \nabla H_0$ is not assumed globally injective.

Fix $\rho > 0$. Work on the complex covering space of the angles:

$$S_\rho := \{q \in \mathbb{C}^n : \|\text{Im } q\|_\infty \leq \rho\}, \quad \mathcal{D}_\rho := G \times S_\rho.$$

The strip S_ρ has unbounded real directions. The physical phase space is

$$\mathbb{T}^n := \mathbb{R}^n / (2\pi\mathbb{Z})^n, \quad \mathcal{P} := G_{\mathbb{R}} \times \mathbb{T}^n = \overline{U} \times \mathbb{T}^n.$$

Write $\pi(p, q) = (p, [q])$ for the real phase projection. On $\mathbb{R}^n \times \mathbb{T}^n$, let μ be the product of action Lebesgue measure and angle measure induced by a 2π -periodic fundamental cell. Thus

$$0 < \mu(\mathcal{P}) = (2\pi)^n m_n(G_{\mathbb{R}}) = (2\pi)^n m_n(U) < \infty.$$

2 Admissible perturbations and the main theorem

A function $H_1 : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ is an admissible perturbation if:

(P1) **Analyticity:** it is jointly holomorphic on an open neighborhood of $\mathcal{D}_\rho$.

(P2) **Periodicity:** for every $(p, q) \in \mathcal{D}_\rho$ and $k \in \mathbb{Z}^n$,

$$H_1(p, q + 2\pi k) = H_1(p, q).$$

(P3) **Conjugation compatibility:** for every $(p, q) \in \mathcal{D}_\rho$,

$$H_1(\bar{p}, \bar{q}) = \overline{H_1(p, q)}.$$

In particular, H_0 and H_1 are real valued on their respective real domains; this follows from the stated conjugation identities.

(P4) **Boundedness:** it is bounded on $\mathcal{D}_\rho$. Its domain norm is

$$\|H_1\|_{G,\rho} := \sup_{(p,q) \in \mathcal{D}_\rho} |H_1(p, q)| < \infty.$$

The total functions above mirror the Lean representation. Analyticity and the other domain properties are required only on the specified sets.

Theorem (the current `theorem1` interface)

For fixed data satisfying Section 1, and for every $\kappa > 0$, there exists $M > 0$, chosen before the perturbation, such that *every* admissible H_1 with

$$\|H_1\|_{G,\rho} \leq M \tag{3}$$

has the following properties. Set

$$H(p, q) := H_0(p) + H_1(p, q), \quad X_H(p, q) := (-\partial_q H(p, q), \partial_p H(p, q)).$$

There exist r with $0 < r \leq \rho$, sets $F_{\text{good}}, F_{\text{bad}} \subset \mathcal{P}$, and a family $\mathcal{T}$ of actual subsets of $\mathcal{P}$, satisfying:

(C1) **Partition and measure.** The good set is compact, both sets are measurable, and

$$\mathcal{P} = F_{\text{good}} \dot{\cup} F_{\text{bad}}, \quad F_{\text{good}} \neq \emptyset,$$

$$\mu(F_{\text{good}}) > 0, \quad (1 - \kappa)_+ \mu(\mathcal{P}) < \mu(F_{\text{good}}), \quad \mu(F_{\text{bad}}) < \kappa \mu(\mathcal{P}),$$

where $a_+ = \max\{a, 0\}$. This is the real-valued reading of the code's `ENNReal.ofReal` inequality. For $0 < \kappa < 1$, the lower bound is $(1 - \kappa)\mu(\mathcal{P})$.

(C2) **Family of tori.** Distinct members of $\mathcal{T}$ are disjoint, and $F_{\text{good}} = \bigcup_{T \in \mathcal{T}} T$. Every $T \in \mathcal{T}$ has the common-width torus realization specified in Section 3.

(C3) **Global trajectories.** For each $z \in F_{\text{good}}$, there is a real-valued lifted trajectory $\gamma : \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{C}^n$ with $\pi(\gamma(0)) = z$, $\gamma'(t) = X_H(\gamma(t))$, and $\pi(\gamma(t)) \in F_{\text{good}}$ for every $t \in \mathbb{R}$.

3 Realization of each invariant torus

For every $T \in \mathcal{T}$, there exist a center $c_T \in G_{\mathbb{R}}$, a frequency $\omega_T \in \mathbb{R}^n$, a lift $L_T = (a_T, b_T) : \mathbb{C}^n \rightarrow \mathbb{C}^n \times \mathbb{C}^n$, an embedding $e_T : \mathbb{T}^n \rightarrow \mathbb{R}^n \times \mathbb{T}^n$, and a map $J_T : \mathbb{C}^n \rightarrow \mathbb{C}^n$, with all of the following properties.

(T1) **Frequency and center.**

$$\nabla H_0(c_T) = \omega_T, \quad k \cdot \omega_T \neq 0 \quad (k \in \mathbb{Z}^n \setminus \{0\}).$$

The family is indexed by its actual torus sets. Distinct tori are allowed to have the same frequency.

(T2) **Joint angle analyticity, periodicity, and reality.** The map L_T is holomorphic on a neighborhood of S_r , and for $q \in S_r$, $k \in \mathbb{Z}^n$,

$$L_T(q + 2\pi k) = (a_T(q), b_T(q) + 2\pi k).$$

Moreover, $L_T(q) \in \mathbb{R}^n \times \mathbb{R}^n$ for every $q \in \mathbb{R}^n$. Analyticity here is joint in all complex angle coordinates for each fixed torus. No regularity in an open set of torus labels is asserted.

(T3) **Embedding and immersion.** The same lift induces the same torus:

$$e_T([q]) = \pi(L_T(q)) \quad (q \in \mathbb{R}^n), \quad e_T(\mathbb{T}^n) = T.$$

The map e_T is a closed topological embedding, and the complex derivative $DL_T(q)$ is injective at every $q \in S_r$. Thus the lift supplies an analytic embedded real n -torus.

(T4) **Angle inverse.** The angle component $Q \mapsto (e_T(Q))_q$ is a homeomorphism of $\mathbb{T}^n$. The map J_T is holomorphic near every real point, and

$$J_T(b_T(q)) = q \quad (q \in \mathbb{R}^n), \quad b_T(J_T(Q)) = Q \quad (Q \in \mathbb{R}^n).$$

These are the inverse identities recorded by the interface; no common complex strip for J_T is required.

(T5) **Displacement.** For every $q \in S_r$,

$$\|a_T(q) - c_T\|_{\infty} < \kappa, \quad \|b_T(q) - q\|_{\infty} < \kappa.$$

In particular, on real angles the parametrization has the familiar form $p = c_T + f_T(q)$, $[q_{\text{physical}}] = [q + g_T(q)]$, with periodic analytic corrections $f_T = a_T - c_T$ and $g_T = b_T - \text{id}$.

(T6) **Original Hamiltonian dynamics for all time.** For every $q \in \mathbb{R}^n$, the real-valued curve

$$\gamma_{T,q}(t) := L_T(q + t\omega_T) \quad (t \in \mathbb{R})$$

satisfies, for every real t ,

$$\frac{d}{dt}\gamma_{T,q}(t) = X_{H_0+H_1}(\gamma_{T,q}(t)), \quad \pi(\gamma_{T,q}(t)) \in T.$$

Consequently the induced motion on T is conjugate via e_T to $[q] \mapsto [q + t\omega_T]$. This expresses invariance and quasi-periodic motion through the explicit orbit equation used by the code.

4 Source alignment and changes from the earlier draft

Version used. Local checkout: `publication/arnold1963-kam`. Source commit:

`f3c70ad77a0005331aca2a244f8f336677fdd454`.

Namespace: `KamProject.Arno1d1963`.

Source paths below are relative to `KamProject/Arno1d1963/`.

English specification	Lean declaration and source file
Section 1 fixed data	<code>GlobalHamiltonianData</code> in <code>Geometry/LocalizationCover.lean</code>
Perturbations and norm	<code>AnalyticPhaseFunction</code> , <code>uniformNorm</code> in <code>Basic/Functions.lean</code>
Threshold and global output	<code>theorem1</code> , <code>Theorem1Result</code> in <code>Main/Theorem1.lean</code>
Individual torus data	<code>KAMTorus</code> in <code>Main/TorusResult.lean</code>
Global family and disjointness	<code>FiniteLocalization.tori</code> and <code>tori_pairwise_disjoint</code> in <code>Tori/CrossChart.lean</code>

Changes to the mathematical statement.

1. Replace the earlier complex-neighborhood condition on all of $\bar{U}$ by (1), including the exact real-slice identity. Impose Hessian nondegeneracy on U . Use $\bar{U} \times \mathbb{T}^n$ for the output partition and $c_T \in \bar{U}$ for the center condition.
2. State conjugation identities explicitly. Real-valuedness on the real slice is a consequence, without an unsupported converse on other complex components. Define the complex domain on the angle covering space, as in `phaseDomain`, and include boundedness explicitly.
3. Put the quantifiers in the order $\forall \kappa > 0 \exists M > 0 \forall H_1$, and use $\|H_1\|_{G,\rho} \leq M$. The threshold depends only on the fixed unperturbed data, ρ , and κ . The companion `theorem1_of_pointwise` accepts $|H_1(z)| < M$ at every domain point; its proof uses the resulting norm bound $\|H_1\|_{G,\rho} \leq M$.
4. Replace the frequency-indexed “foliation” by a family of actual torus sets. Retain the common positive width, compactness, positive measure, both strict measure bounds, and the nonempty good set supplied by the code.
5. Express all torus properties using the same lift, embedding, center, frequency, and angle inverse. Keep the displacement bound on S_r and the orbit equation for the original Hamiltonian at every real time.

Relation to the literature. This is a mathematical rendering of the specified Lean interface. It does not assert equivalence with the full range of hypotheses in Arnold’s original paper. Its classical reference is V. I. Arnol’d, *Proof of a theorem of A. N. Kolmogorov on the invariance of quasi-periodic motions under small perturbations of the Hamiltonian*, Russian Mathematical Surveys **18**(5) (1963), 9–36, especially Section 2.1. This revision aligns the stated interface; it is not a new audit of the entire formal proof.